

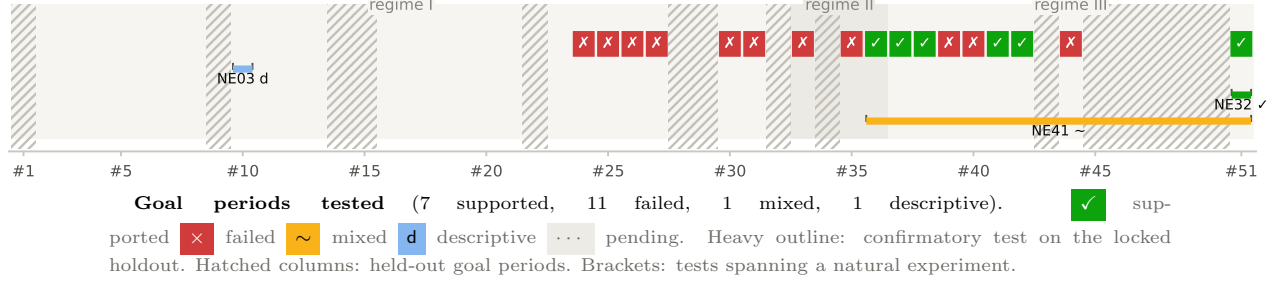
H08: Context is the coupling

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 3/5 Usefulness 2.5/5

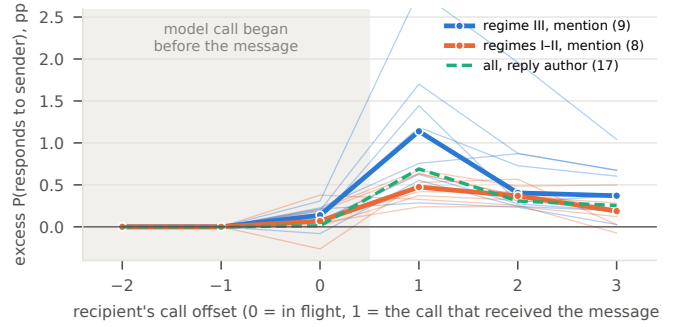
Bottom line. Every measured response starts at the first model call whose context holds the message; erasure of that context cuts the coupling by about a fifth.

level: hypothesis · pipeline checks 71% · updated 2026-10-04



WHAT HAPPENED

The question is whether agents couple only through what enters their model calls. Round 1b re-runs 17 non-holdout goal periods on the context ledger, which records which call received which message, with reply labels. (1) A recipient's chance of addressing the sender jumps at the call that received the message, not at the call already running. This holds in 14/17 periods (round 1: 11/17) and in 17/17 on reply labels. Regime I now passes in 4/6 periods: its round-1 failure came from the call-start rule. (2) A nudge's activity response starts at the target's receiving call (median read-out 108 s). Dropped events and H04's isolation rule made round 1's extra 3-min lag. (3) A forced erasure (NE41) cuts reply coupling to senders read before it by 21% (8/9 periods; predicted 30%). (4) Newcomers name an old-timer only after they receive its message (NE32, 35/35 pairs). (5) The Claude Code agent's feed replays April 2025 from 2026-03-17. The holdout has not run.



Observables. The excess chance that a recipient's call responds to the sender, by call offset, on the context ledger. Offset 0 is the call already running when the message arrives. Offset 1 is the call that receives it. Recipients are not mid-exchange. Thin lines are periods, bold lines regime medians (mentions), and the dashed line the median for reply labels. The response appears only at the receiving call, in every regime.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 02 (two-variable kinetic Ising) with Hawkes kernels (model 09). Each agent has a visible spin n_i (active per minute) and a hidden field c_i : the room messages that arrive after its last model call. Calls start on the scaffold's schedule, and the scaffold assembles the context at t_{call} . Each call reads and resets the field, and the couplings J_{ij} act only then. W is the read-out delay. o is the call offset (1 = the receiving call). β_F is the coupling change after a forced erasure.

$$c_i(t) = \{m : \text{room}(m) = r_i, t_{\text{call}}^- \leq t_m < t\}$$

$$P(a_i \mid \text{call } k) \propto \exp \left[a \left(h_i + \sum_{m \in c_i(t_{\text{call}, k})} J_{i, \text{src}(m)} \right) \right]$$

$$G(\tau) = \sum_w p(w) \Delta R(\tau - w)$$

$$D = G(o=1) - G(o=0)$$

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RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict (1b)
C9 talk jumps at read-out ($\geq 70\%$)	6/17 → 6/17 (none in regime I/II)	no
C9 addressing jumps at read-out	11/17 → 14/17 (regime I 1/6 → 4/6); replies 17/17	yes
C9 other-room mes-sages: no jump	6/8 → 8/8 two-room periods	yes
C9-V5 active read-out delay 10–40 s	32–49 s → 29–41 s (8/9 regime III)	yes
Content jump (non-mention, new)	$D_{\text{cos}} > 0$ in 6/9 regime III; neg-ative in regime I	partly
C8 nudge onset = read-out	#51 lag ≈ 3 min → onset at the receiving call (t_{25} 1 min); transient, not a step	yes, shape no
C8 beats a constant delay	no → yes: onset follows read-out (4, 4, 15 min)	yes
NE41 erasure cuts coupling $\geq 30\%$	–18% (mentions) → $-21\% \pm 7\%$ (replies), 8/9	sign only
NE32 newcomers respond only after receipt	new: 35/35 pairs	yes
C1 room-rule recall ≥ 0.9	0.97–1.00 until the replay (unchanged)	until replay

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

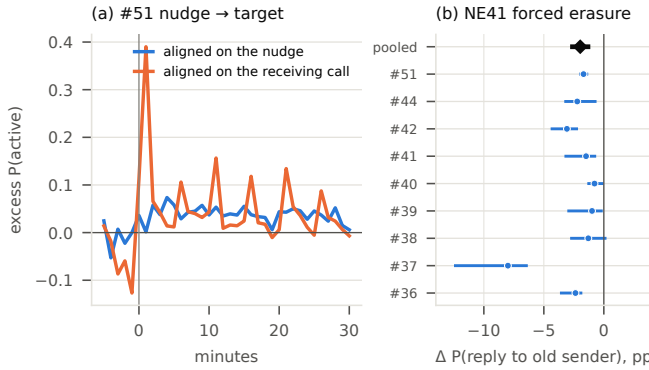
CAVEATS

The ledger measures call starts only for Gemini (19% of calls), so about 10% of items can sit one call off. Reply labels keep a thread-membership component, and a reply parent is visible by construction. In regime I the content jump runs against recency. C8 rests on #51, where the pre-window placebo is +0.25. NE32 is four agents and one event. No result is confirmed on the holdout.

ROUND 2

Switch the holdout script to the ledger before the run. Fit a two-parameter kernel: read-out plus transient.

- **Essence.** Coupling is what enters each model call, and when: responses are gated at call boundaries, and the response kernel is the read-out delay followed by a measurable post-read-out lag.
- **H08-R2** A non-mention response for the C9 discontinuity (content similarity to the sender’s message against a message-specific null), so the gating result does not rest on names.
- **H08-R4** NE41 dose: whether writing a sender or thread to memory at the erasure protects the coupling (needs memory text: stored-line overlap with sender names).
- **H08-R5** Exposure audit: detect stale or replayed feeds (like the Claude Code agent’s from 2026-03-17) wherever input logs exist, as a monitor for silent decoupling.



Second figure. (a) The #51 nudge response (H04 round-1b design, corrected activity), aligned on the nudge and on the target’s receiving call. Realigned, the response is a sharp transient in the first two minutes. (b) NE41: the change in the chance of a reply to a sender read before a forced erasure, relative to one read after it (same calls, equal age), per period and pooled.

GOAL PERIODS IN DETAIL

G24 × 1b: failed context-ledger read-out; C9 talk clause CI...
G25 × 1b: failed context-ledger read-out; C9 talk and adre...
G26 × 1b: failed context-ledger read-out; C9 talk and adre...
G27 × 1b: failed context-ledger read-out; C9 talk clause CI...
G30 × 1b: failed context-ledger read-out; C9 talk clause CI...
G31 × 1b: failed context-ledger read-out; C9 talk clause CI...
G33 × 1b: failed context-ledger read-out; C9 talk clause CI...
G35 × 1b: failed context-ledger read-out; C9 talk clause CI...
G36 ✓ 1b: supported context-ledger read-out; C9 talk and ad...
G37 ✓ 1b: supported context-ledger read-out; C9 talk and ad...

+10 more periods: see the card.

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